

Lab 5 – Homework Solutions

October 21, 2024

A Homework

Homework 5.1. Consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 10^{10} & 1 & 1 \\ 10^{10} & 1 & 1 & 10^{10} \\ 1 & 1 & 10^{-10} & 1 \\ 1 & 10^{10} & 1 & 10^{10} \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 10^{10} + 3 \\ 2 \cdot 10^{10} + 2 \\ 3 + 10^{-10} \\ 2 \cdot 10^{10} + 2 \end{bmatrix}$$

such that the exact solution is $\mathbf{x} = [1 \ 1 \ 1 \ 1]^T$.

- Solve the system using LU decomposition with pivoting
- Compute the infinity norm of the error associated with the solution of the previous item
- Can we apply the Cholesky decomposition to matrix $\mathbb{A}$?
- Is the normalized residual $\|\mathbf{b} - \mathbb{A}\mathbf{x}\|/\|\mathbf{b}\|$ a good estimator for the relative error? Motivate your answer.

Solution Homework 5.1.

hw_5_1.m

```
%% Homework 4.6
clc
clear all
close all

A = [1 1e10 1 1;
     1e10 1 1 1e10;
     1 1 1e-10 1;
     1 1e10 1 1e10];
x = [1 1 1 1]';
b = [1e10 + 3; 2e10 + 2; 3 + 1e-10; 2e10 + 2];

% solution with LU decomposition without pivoting
[L1, U1] = lu_decomposition(A);
y1 = L1 \ b;
x1 = U1 \ y1

% The computed solution is incorrect! The third component is not 1.

% solution with LU decomposition with pivoting
[L2, U2, P2] = lu(A);
y2 = L2 \ (P2*b);
x2 = U2 \ y2
% The computed solution is ok.

%% Compute the infinity norm of the error associated with the solution of the previous item
% infinity norm of the error
err1=norm(x - x1, inf)
err2=norm(x - x2, inf)
% As already noted, the best solution is that obtained with the pivoting.

%% Can we apply the Cholesky decomposition to matrix A?
% Cholesky decomposition? -> no, because A is symmetric but not definite
% positive definite, since the eigenvalues are not all positive.
eigenvalues=eig(A)

% normalized residual
res1=norm(b - A*x1, inf) / norm(b, inf)
res2=norm(b - A*x2, inf) / norm(b, inf)

% In both cases the normalized residuals are very small.
% Nevertheless the relative error is not small in both cases.
% The normalized residual is a good estimate of the relative error
% only when the condition number of the matrix is small!
% Furthermore if the condition number is large the pivoting is necessary to avoid the
amplification of round-off errors.

K=cond(A, inf)
```

Homework 5.2. Compute the condition number and the determinant of the following matrices » $A = \text{diag}(0.1 \cdot \text{ones}(15,1))$;
» $B = \text{triu}(\text{rand}(15),14) \cdot 1e5 + \text{diag}(0.1 \cdot \text{ones}(15,1))$;
Is, in general, the determinant a good measure of the condition number of the matrix?

Solution Homework 5.2.

hw_5_2.m

```
%% Homework 5.2
clc
clear all
close all
%% Compute the condition number and the determinant of the following
```

```
% matrices
A = diag(0.1*ones(15,1));
B = triu(rand(15),14)*1e5 + diag(0.1*ones(15,1));

det(A)
cond(A)

det(B)
cond(B)

% The two matrices have the same determinant but B is very badly
% conditioned. Thus, we cannot say that the determinant is a good indicator
% for the condition number of a matrix.
```

Homework 5.3. Consider the following matrix

$$\mathbb{A} = \begin{bmatrix} 1.001 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1.001 & 1 & 1 \\ 1 & 1.001 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1.001 \end{bmatrix}.$$

- Compute the condition number $\mathcal{K}_{\infty}(\mathbb{A})$ using MATLAB command `cond`.
- Then compute $\mathcal{K}_{\infty}(\mathbb{A})$ using its definition.
- Finally estimate $\mathcal{K}_{\infty}(\mathbb{A})$ by **perturbing a suitable linear system** having $\mathbb{A}$ as coefficient matrix.

Solution Homework 5.3.

hw_5_3.m

```
%% Homework 5.3
clc
clear all
close all

A = ones(5,5);
A(1,1) = 1.001;
A(3,2) = 1.001;
A(2,3) = 1.001;
A(5,5) = 1.001;
A
%%Compute the condition number K_inf(A) using MATLAB command cond
KA = cond(A, inf)
%% Then compute K_inf(A) using its definition.
KA_def = norm(A, inf) * norm(inv(A), inf)

%% Finally estimate K (A) by perturbing a suitable linear system
% having A as coefficient matrix
n_trials = 100;
KA_store = zeros(n_trials,1);
pert_mag = 1e-6; % Perturbation magnitude
x = ones(5,1);
b = A*x;

for (i=1:n_trials)
    deltab = pert_mag * randn(5,1);
    deltax = A\deltab;
    KA = norm(deltax,inf)/norm(x,inf) * norm(b,inf)/norm(deltab,inf);
    KA_store(i) = KA;
end
KA_est = max(KA_store)

figure
set(gcf, 'Position', [100 100 1000 500])
plot([1:n_trials], KA_store, 'bx-', 'LineWidth', 2, 'MarkerSize', 8)
```

```

hold on, box on
plot([1:n_trials], cond(A,inf)*ones(n_trials,1), 'r--', 'LineWidth',2)
axis([0 n_trials+1 0 1.1*cond(A,inf)])
set(gca, 'LineWidth',1.5)
set(gca, 'FontSize',16)
xlabel('trials')
ylabel('K_{est}(A)')

```

Homework 5.4. Consider the linear systems $\mathbb{A}_i \mathbf{x} = \mathbf{b}_i$, $i = 1, \dots, 4$ with

$$\mathbb{A}_1 = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} \quad \mathbb{A}_2 = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$\mathbb{A}_3 = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} \quad \mathbb{A}_4 = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

with $\mathbf{x} = [1 \ 2 \ 3]^T$ and $\mathbf{b}_i = \mathbb{A}_i \mathbf{x}$.

- Write a MATLAB function that implements the generic iterative scheme $\mathbf{x}^{(k+1)} = \mathbb{B}\mathbf{x}^{(k)} + \mathbf{g}$. Use the stopping criterion based on the increment
- Study the convergence for Jacobi and Gauss-Seidel methods. Check the results using the function at the previous item, setting $\mathbf{x}_0 = [0, 0, 0]^T$, $\text{tol} = 1.e-6$, $\text{maxit} = 100$

Solution Homework 5.4.

hw_5_4.m

```

%% Exercise 5.3
clc
clear all
close all

% Matrix definition. Uncomment the one you want to analyze
A = [1 2 2; 1 1 1; 2 2 1]; % None of the methods converge
%A = [2 1 -2; 1 2 1; 2 1 2]; % Both methods converge
%A = [1 2 -2; 1 1 1; 2 2 1]; % Just Jacobi method converges
%A = [1 1 -2; 1 2 1; 2 1 2]; % Just Gauss-Seidel method converges

x = [1 2 3]'; % Exact solution
b = A*x; % Right hand side

% Iteration matrices.
D = diag( diag(A) );
E = -tril(A, -1);
% L = tril(A, -1);
% U = triu(A, 1);

% Bj = -D \ (L+U) % Jacobi method
% Bgs = -(D+L) \ U % Gauss-Seidel method

% Bj = inv(D)*(D-A); the following implementation is faster as declared by
% Matlab
Bj = D \ (D-A)
% Bgs = inv(D-E)*(D-E-A); the following implementation is faster as declared by
% Matlab
Bgs = (D-E) \ (D-E-A)

% The matrices do not have particular properties like being strictly diagonally dominant or
symmetric and positive definite.

norm(Bj)
norm(Bgs)

% Moreover, the 1,2 and infinity norms of Bj and Bgs for all matrices A are
% ALL bigger than 1. So we cannot use this argument to conclude convergence
% for any case.
% Hence we have to compute the spectral radii of both matrices to determine
% whether the methods are convergent or not.

```

```

rhoBj = max( abs( eig(Bj) ) )
rhoBgs = max( abs( eig(Bgs) ) )

% Iteration vector
gj = D \ b;
ggs = (D-E) \ b;

x0 = [0 0 0]';
tol = 1e-6;
maxit = 100;

% Jacobi method
[xj, iterj, incrl] = stationary_method(Bj, gj, x0, tol, maxit);
xj
iterj
% Gauss-Seidel method
[xgs, itergs, incrgs] = stationary_method(Bgs, ggs, x0, tol, maxit);
xgs
itergs
%{
Convergence of Jacobi and Gauss-Seidel method are not related to each other.
As we could see they can both diverge or be convergent or also just one can
converge.
It always depends on the matrix. They both have some advantages. Since in Jacobi
method each component of the approximation of the solution is obtained
using information at the previous iteration, the algorithm is highly
parallelizable. This is not possible for Gauss-Seidel method, since each
component is obtained using the information on the other components
already computed at the current iteration. However this smarter use of
the information, can speed up convergence.
%}

```

stationary_method.m

```

function [x, iter, incr] = stationary_method(B, g, x0, tol, max_it)
% stationary_method Solve a linear system with a stationary iterative method of the form
%  $x^{(k+1)} = B \cdot x^{(k)} + g$ 
%
% [x, iter, incr] = stationary_method(B, g, x0, tol, max_it)
%
% Inputs : B    = iteration matrix
%          g    = iteration vector
%          x0   = starting point
%          tol  = requested tolerance
%          max_it = maximum number of allowed iterations
% Outputs : x   = approximation of the solution
%          iter = performed iterations
%          incr = last increment (can be used to estimate the error)

% Initialization
iter = 0;
incr = inf;
x = x0;

while (incr > tol) && (iter < max_it)

    iter = iter+1;
    x_old = x;           % Save the previous approximation

    x = B*x + g; %xnew = B*xold + g

    incr = norm(x - x_old, 2); % 2-norm of the increment,
    %you can choose the residual
    % res = norm(b - A*x, 2);
end

end

```

Homework 5.5. Let the 10×10 square matrix $\mathbf{A}$ be

$$\mathbf{A} = \begin{bmatrix} 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 3 \end{bmatrix}$$

Consider the linear system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{x} = [1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]^T$.

1. Study the convergence of the Jacobi and Gauss-Seidel methods by investigating the relation between spectral radius of the two iteration matrices
2. Check the results using the functions implemented in 5.3 and choosing $\mathbf{x}_0 = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T$, $\text{tol} = 1.e-12$, $\text{maxit} = 500$

Solution Homework 5.5.

hw_5_5.m

```
%% Exercise 5.4
clc
clear all
close all

% Matrix definition.
n = 10;
A = 3* eye ( n ) - 2* diag ( ones (n -1 ,1) ,1) - diag ( ones (n -1 ,1) , -1);

% Calculation of the b vector
b = A * ones ( n ,1);

% Iteration matrices.
D = diag( diag(A) );
E = -tril(A, -1);

% Bj= inv(D)*(D-A); the following implementation is faster as declared by
% Matlab
Bj = D \ (D-A)
% Bgs = inv(D-E)*(D-E-A); the following implementation is faster as declared by
% Matlab
Bgs = (D-E) \ (D-E-A)

% The matrices do not have particular properties like being strictly diagonally dominant or
symmetric and positive definite.
norm(Bj)
norm(Bgs)

% Moreover, the 1,2 and infinity norms of Bj and Bgs for all matrices A are
% ALL bigger than 1. So we cannot use this argument to conclude convergence
% for any case.
% Hence we have to compute the spectral radii of both matrices to determine
% whether the methods are convergent or not.

rhoBj = max( abs( eig(Bj) ) )
rhoBgs = max( abs( eig(Bgs) ) )

% Iteration vector
gj = D \ b;
ggs = (D-E) \ b;

x0 = zeros(10,1);
tol = 1e-12;
maxit = 500;

% Jacobi method
[xj, iterj, incrj] = stationary_method(Bj, gj, x0, tol, maxit)
```

```

% Gauss-Seidel method
[xgs, itergs, incrgs] = stationary_method(Bgs, ggs, x0, tol, maxit)

stationary_method.m

function [x, iter, incr] = stationary_method(B, g, x0, tol, max_it)
% stationary_method Solve a linear system with a stationary iterative method of the form
%  $x^{(k+1)} = Bx^{(k)} + g$ 
%
% [x, iter, incr] = stationary_method(B, g, x0, tol, max_it)
%
% Inputs : B = iteration matrix
%          g = iteration vector
%          x0 = starting point
%          tol = requested tolerance
%          max_it = maximum number of allowed iterations
% Outputs : x = approximation of the solution
%           iter = performed iterations
%           incr = last increment (can be used to estimate the error)

% Initialization
iter = 0;
incr = inf;
x = x0;

while (incr > tol) && (iter < max_it)

    iter = iter+1;
    x_old = x; % Save the previous approximation

    x = B*x + g; %xnew = B*xold + g

    incr = norm(x - x_old, 2); % 2-norm of the increment,
    %you can choose the residual
    % res = norm(b - A*x, 2);
end

end

```

Homework 5.6. Consider the linear system $\mathbb{A}_3 \mathbf{x} = \mathbf{b}$ with

$$\mathbb{A}_3 = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b}_3 = \mathbb{A}_3 \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

- Apply Jacobi method to compute the solution with a tolerance of 10^{-5} and 10^{-8} . What do you observe?
- Compute the iteration matrix $\mathbb{B}_J$. Evaluate its spectral radius $\rho(\mathbb{B}_J)$ and $(\mathbb{B}_J)^3$. Relying on the results, motivate what you found at the previous point.

Solution Homework 5.6.

hw_5_6.m

```

%% Homework 5.4
clc
clear all
close all

A = [1 2 -2;
     1 1 1;
     2 2 1];
b = A * [1 2 3]';
% Apply Jacobi method to compute the solution with a tolerance of 105 and 108
D = diag(diag(A));

```

```

L = tril(A, -1);
U = triu(A, 1);
Bj = -D \ (L+U);
gj = D \ b;

x0 = zeros(3,1);

iter = 0;
%tol = 1e-5;
tol = 1e-8;
maxit = 100;

[xj, iterj, incrj] = stationary_method(Bj, gj, x0, tol, maxit)

% Four iterations are performed in both cases, independently on the tolerance values we
considered.
% Notice that the last increment is actually zero!

% The spectral radius is very small.

eig(Bj)
rhoBj = max( abs( eig(Bj) ) )

% Indeed, computing by hand the eigenvalues we see that
% lambda_1 = lambda_2 = lambda_3 = 0, so the spectral radius is actually 0.

% Compute the cube of Bj
Bj^2
Bj^3      % At the third step we get the null matrix.

% Thus, for all k > 3
% Jacobi method has converged to the exact solution

```

Homework 5.7. Consider a linear system whose coefficient matrix is obtained with the commands » $B = \text{rand}(5) + \text{diag}(10 \cdot \text{ones}(5,1))$; » $A = B \cdot B'$; Analyze the convergence of Jacobi and Gauss-Seidel methods.

Solution Homework 5.7.

hw_5_7.m

```

%% Homework 5.6

clc; close all; clear
n=5;
B=rand(n)+ diag(10*ones(n,1));
A=B*B';
% Iteration matrices.
D = diag( diag(A) );
E=-tril(A,-1);
F=-triu(A,1);

Bgs=(D-E)\F;      %Jacobi method
Bj = eye(n) - D\A; %Gauss-Seidel method

% Compute the spectral radii of both matrices to determine whether the
% methods are convergent or not.

rhoBj = max( abs( eig(Bj) ) )
rhoBgs = max( abs( eig(Bgs) ) )

% It can be noted that both spectral radii are less than 1, so that
% both method are convergent.

%test
x=ones(n,1);
b=A*x;

```



```

% Iteration vector
gj = D \ b;
ggs = (D-E) \ b;

x0 = zeros(n, 1);
tol = 1e-12;
maxit = 1000;

% Jacobi method
[xj, iterj, incrj] = stationary_method(Bj, gj, x0, tol, maxit)

% Gauss-Seidel method
[xgs, itergs, incrgs] = stationary_method(Bgs, ggs, x0, tol, maxit)

```

Homework 5.8. Consider a system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 3 & -1 & 0 & 0 & 0 & -1 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 3 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & 0 \\ 0 & -1 & 0 & -1 & 3 & -1 \\ -1 & 0 & 0 & 0 & -1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

- Study the convergence of Jacobi and Gauss-Seidel methods if applied to the system $\mathbb{A}\mathbf{x} = \mathbf{b}$.
- Compute the spectral radii of the iteration matrices of both methods.
- How many iterations are needed to obtain the solution with a tolerance of 10^{-8} starting from $\mathbf{x}^{(0)} = \mathbf{0}$?

Solution Homework 5.8.

hw_5_8.m

```

%% Homework 5.7
clc
clear all
close all

A = diag(3*ones(6,1)) - diag(ones(5,1), 1) - diag(ones(5,1), -1);
A(1, 6) = -1;
A(2, 5) = -1;
A(5, 2) = -1;
A(6, 1) = -1;
A

b = [1 0 1 1 0 1]';

% The matrix is clearly symmetric, and the computation of the eigenvalues
% shows that is positive definite, so the Gauss-Seidel method converges.
% We cannot conclude anything on the convergence of the Jacobi method because
% the matrix is only weakly diagonally dominant.

D = diag(diag(A));
L = tril(A, -1);
U = triu(A, 1);

Bj = -D \ (L+U);
rhoBj = max(abs(eig(Bj)))

Bgs = -(D + L) \ U;
rhoBgs = max(abs(eig(Bgs)))

% Both Jacobi and Gauss-Seidel method will converge.

```

```

x0 = zeros(6,1);
tol = 1e-8;

% Jacobi
normBj = norm(Bj, 2)
% Choose 2-norm because the implementation of stationary_method provides
% stopping criterion in 2-norm. This is actually the spectral radius itself,
% since matrix A is symmetric.
gj = D \ b;
xlj = Bj*x0 + gj;
kjmin_2 = log(tol*(1 - normBj)/norm(xlj-x0)) / log(normBj)

% Rounding up to the first integer the estimate says that tolerance
% will be satisfied in no more than 91 iterations.

% Gauss-Seidel
normBgs = norm(Bgs)
ggs = (D + L) \ b;
xlgs = Bgs*x0 + ggs;
kgsmin_2 = log(tol*(1 - normBgs)/norm(xlgs-x0)) / log(normBgs)

% The estimate says that tolerance will be satisfied in no more than 89 iterations

[xj,niterj] = stationary_method(Bj, gj, x0, tol, 1000)
[xgs,nitergs] = stationary_method(Bgs, ggs, x0, tol, 1000)

% We get convergence in respectively 83 and 45 iteration, in accordance
% with what we found from the estimates.

```

Homework 5.9. Consider a system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 50 & 1 & 2 \\ 1 & 5 & 2 \\ 2 & 2 & 7 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

- Compute $\|\mathbb{A}\|_{\infty}$ and $\|\mathbb{A}\|_1$.
- Is $\mathbb{A}$ symmetric and positive definite?
- Compute the LU decomposition using `lu_decomposition`.
- Compute $\mathcal{K}_2(\mathbb{A})$. Is $\mathbb{A}$ well-conditioned?
- Which methods (direct and iterative) can be employed for solving the system? Motivate. If Jacobi method is in that list, estimate both the convergence rate and the error after 4 iterations starting with $\mathbf{x}^{(0)} = \mathbf{0}$. Numerically validate the results.

Solution Homework 5.9.

hw_5_9.m

```

%% Homework 5.8
clc
clear all
close all

A = [50 1 2;
     1 5 2;
     2 2 7];
b = [1 0 0]';

norm(A,1)
norm(A,inf)

% Yes, since A is symmetric and strictly diagonally dominant, it is
% positive definite. We can also check with the eigenvalues: all positive.
eig(A)

```

```

[L1, U1] = lu_decomposition(A)

% The condition number is quite small, so the matrix is well conditioned.
cond(A,2)

% All the methods we saw are suitable for the solution of the linear system,
% since the matrix is symmetric and positive definite.

D = diag(diag(A));
L = tril(A, -1);
U = triu(A, 1);
Bj = -D \ (L+U);
rhoBj = max(eig(Bj))

% Asymptotic convergence rate
rate_jacobi = -log10(rhoBj)

% We now compute the estimate of the error after 4 iterations
x0 = zeros(3,1);
x1 = Bj*x0 + D\b;

esterr4 = norm(Bj)^4/(1-norm(Bj))*norm(x1-x0)

% We now compute the approximated solution after four steps.
xj = x0;
for i=1:4
    xj = Bj*xj + D\b;
end
x4 = xj
xtrue = A\b;

norm(xtrue-x4)
esterr4

% The error is lower than the estimate as we expected it to be.

```

Homework 5.10 (Exam February 14, 2012). Let $\mathbb{A} \in \mathbb{R}^{n \times n}$, $n > 3$ be defined as

$$\mathbb{A} = \begin{bmatrix} 1 & 0 & \cdots & 0 & (1/\alpha) \\ 0 & 1 & & & 0 \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}, \quad 0 < \alpha < 1.$$

- Write the iteration matrix $\mathbb{J}$ for the Jacobi method applied to the solution of the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ and compute $\mathbb{J}^2$ and $\mathbb{J}^3$.
- Let $\alpha = 10^{-4}$, $\mathbf{x}^{(0)} = \mathbf{b} = [1, 1, \dots, 1]^T$. Will the Jacobi method converge? How many iterations will be required to ensure that the infinity norm of the error is less than 10^{-9} ?
- Let $\mathbb{B} = \mathbb{A}^T \mathbb{A}$ and consider the linear system $\mathbb{B}\mathbf{x} = \mathbf{b}$. Compute the iteration matrix $\mathbb{G}$ for the Gauss-Seidel method. Will the Gauss-Seidel method converge?
- Repeat point b applying Gauss-Seidel method.

Solution Homework 5.10.

hw_5_10.m

```

%% Homework 5.9
clc; close all; clear

n=5;
alpha=1e-4;
A= eye(n);

```

```

A(1,n)= 1/alpha;

A

%% Write the iteration matrix J for the Jacobi method applied to the
% solution of the linear system Ax = b and compute J^2 and J^3

D = diag(diag(A));
L = tril(A, -1);
U = triu(A, 1);

J = -D \ (L+U);
J2=J*J;
J3=J2*J;
%J2 is the null matrix. % Thus, for all k > 2
% Jacobi method has converged to the exact solution

%% Let x0=b=ones(n,1). Will the Jacobi method
% converge? How many iterations will be required to ensure that the
% infinity norm of the error is less than 1e-9
rhoJ = max(abs(eig(J)))
% Jacobi method will converge because rhoJ<1

x0=ones(n,1);
b=x0;
tol=1e-9;
normJ= norm(J, 2)
% Choose 2-norm because the implementation of stationary_method provides
% stopping criterion in 2-norm. This is actually the spectral radius itself,
% since matrix A is symmetric.
gj = D \ b;
x1j = J*x0 + gj;
kjmin = log(tol*(1 - normJ)/norm(x1j-x0)) / log(normJ)

% The estimate says that tolerance will be satisfied in no more than 85 iterations

[xj,niterj] = stationary_method(J, gj, x0, tol, 1000)
% Jacobi converges in two iterations as expected

% Let B= A'A. Compute iteration matrix G for the Gauss-Seidel method. Will the Gauss-Seidel
% method converge?

B=A'*A;
G = -(D + L) \ U;
rhoBgs = max(abs(eig(G)))
% GS method will converge because rhoJ<1

% Gauss-Seidel
normG = norm(G)
ggs = (D + L) \ b;
x1gs = G*x0 + ggs;
kgsmin = log(tol*(1 - normG)/norm(x1gs-x0)) / log(normG)

% The estimate says that tolerance will be satisfied in no more than 89 iterations

[xgs,nitergs] = stationary_method(G, ggs, x0, tol, 1000)

```